

Development of ADA4870 Simulator

for the Output Channel of a Three-Phase High-Frequency Generator

(with ADA4807-2 Preamplifiers)

Technical Note TN-2024-01

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Introduction and Objectives

Purpose of the Circuit

This is an engineering analysis of the schematic `ada4807_4870.asc` — a simulation of the output stage based on the ADA4870 operational amplifier with ADA4807-2 current preamplifiers. The circuit is designed to amplify the current-output signal from the AD9705 DAC and drive a low-impedance load.

Key characteristics of the ADA4870 amplifier [1]:

- Unity-gain bandwidth: 52 MHz (large signal), 70 MHz (small signal)
- Slew rate: 2500 V/ μ s
- Maximum output current: 1 A
- Supply voltage: from 10 V to 40 V (single or dual supply)
- Quiescent current: 32.5 mA (typical)

Key characteristics of the ADA4807-2 [2]:

- Low-noise rail-to-rail operational amplifier
- Bandwidth 180 MHz (gain +1)
- Slew rate 225 V/ μ s
- Low noise: 3.1 nV/ $\sqrt{\text{Hz}}$
- Supply range: 2.7 V ... 10.5 V

Project File Structure

Updated: after refactoring, the project is structured as the package `adasim/` with separation into `core/` (pure component calculation), `ltspice_io/` (parsing `.asc/.asy`, running LTspice), and `report/` (plots and text report). The full structure and purpose of each module are described in the project's `README.md` (section "Architecture"), and are not duplicated here to avoid divergence with future changes. Key files:

- `ada4807_4870.asc` — LTspice simulation schematic
- `config.yaml` — configuration (was `config.json` in earlier versions)
- `out/report.txt`, `out/*.png` — text report and plots from the run
- `net/*_readable.txt` — readable circuit report with pin names (generated directly from `.asc`, not via intermediate `.net`)

The circuit looks as follows:

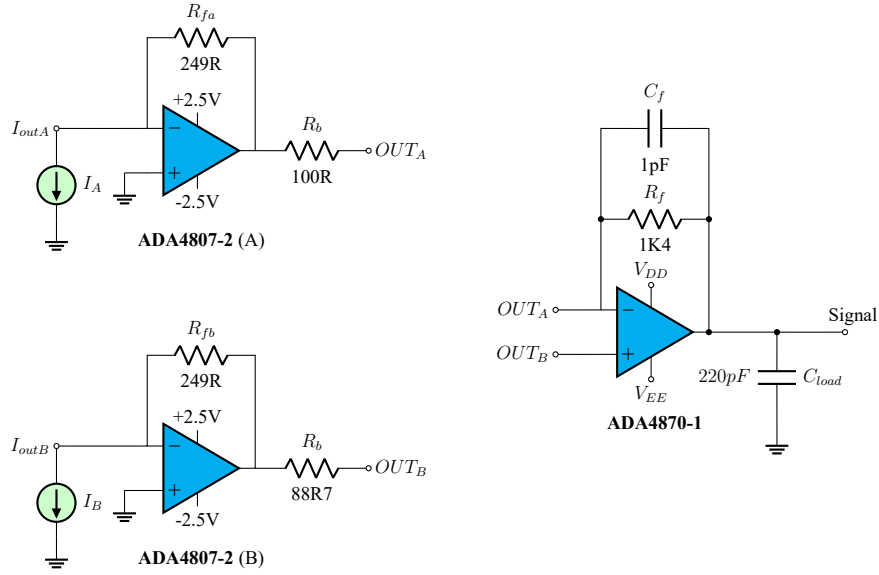


Рис. 1: Overall connection diagram of ADA4807-2 to ADA4870-1

- The differential pair I_{outA} , I_{outB} of the AD9705 are represented by two current sources.
- The current sources generate about 4 mA (the maximum output current of the AD9705 per datasheet does not exceed 5 mA)
- Parasitic parameters of the current sources:
 - Output capacitance of channels I_{outA} , I_{outB} — 5 pF
 - Output resistance of channels I_{outA} , I_{outB} — 150 k Ω
- The signal of the first source is defined by:

$$I = \frac{I_{FS}}{2} + \frac{I_{FS}}{2} (\sin 2\pi \cdot \omega \cdot t) \quad (1)$$

- The signal of the second source is defined by (shifted by π):

$$I = \frac{I_{FS}}{2} + \frac{I_{FS}}{2} (\sin 2\pi \cdot \omega \cdot t + \pi) \quad (2)$$

- ω is the frequency. In this case, 1 MHz
- $V_{DD} = 18V$,
- $V_{EE} = -18V$

In LTspice the schematic looks like:

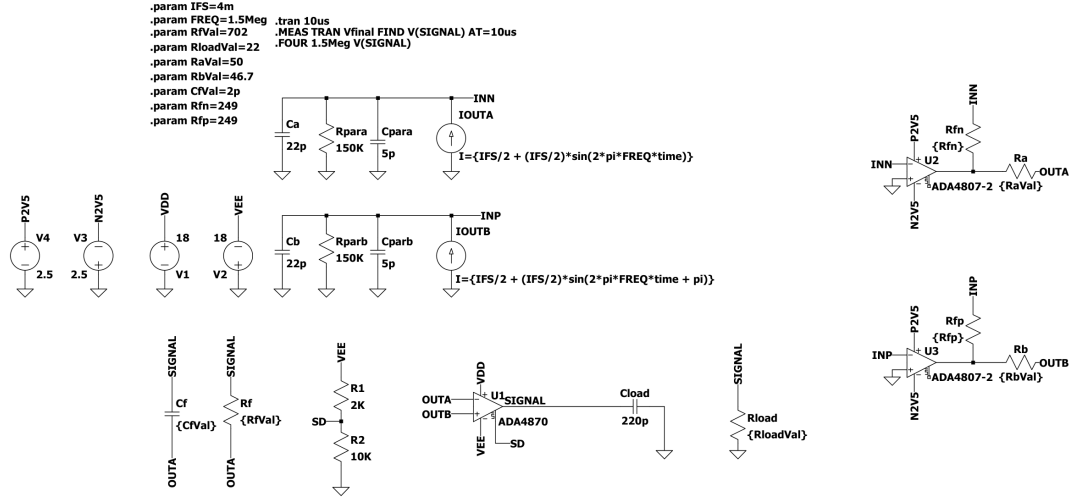


Рис. 2: LTspice simulation schematic

Circuit Description

Block Diagram

The circuit consists of the following functional blocks:

1. Current sources I_{OUTA} and I_{OUTB} — models of the DAC AD9705 current outputs
2. Input transimpedance amplifiers (TIA) based on the dual ADA4807-2 (U2, U3), converting current to voltage
3. ADA4870 input network: resistors R_a , R_b and parasitic elements R_{para}/C_{para} (models of DAC output impedance)
4. ADA4870 differential amplifier (U1) with feedback network R_f/C_f
5. Load network: R_{Load} and C_{Load}
6. Voltage divider R_1/R_2 for SD input control

Current Sources (I_{OUTA} , I_{OUTB})

The sources model the differential current output of the AD9705 DAC [3]. The currents vary in antiphase:

$$I_{OUTA} = \frac{I_{FS}}{2} + \frac{I_{FS}}{2} \sin(2\pi ft), \quad (3)$$

$$I_{OUTB} = \frac{I_{FS}}{2} + \frac{I_{FS}}{2} \sin(2\pi ft + \pi), \quad (4)$$

where I_{FS} is the DAC full-scale current (in the current simulation $I_{FS} = 4$ mA, which corresponds to reduced output current operation), $f = 1$ MHz is the signal frequency.

Each source is unidirectional: the current varies from 0 to I_{FS} , the DC component $I_{FS}/2$ is cancelled by the differential configuration. In the simulation, BI sources (LTspice behavioral current source) are used.

Transimpedance Amplifiers (ADA4807-2)

Two channels of the dual ADA4807-2 (U2, U3) operate as current-to-voltage converters (TIA). Each channel connection:

- Non-inverting input connected to ground (GND).
- Inverting input connected to the DAC current output (I_{OUTA} for U2, I_{OUTB} for U3).
- Feedback resistor R_3 (249 Ω) connected between output and inverting input (for U2); R_5 (249 Ω) for U3.

The virtual ground at the inverting input ensures that the input current flows through the feedback resistor, generating the output voltage:

$$V_{N001} = -I_{OUTA} \cdot R_3, \quad (5)$$

$$V_{N002} = -I_{OUTB} \cdot R_5. \quad (6)$$

The differential voltage at the ADA4870 inputs (ASA and ASB) is

$$V_{\text{diff}} = V_{N001} - V_{N002} = -(I_{OUTA} - I_{OUTB}) \cdot R_3 = -I_{FS} \sin(2\pi ft) \cdot 249 \Omega, \quad (7)$$

since $I_{OUTA} - I_{OUTB} = I_{FS} \sin(2\pi ft)$. The amplitude of the differential signal is

$$V_{\text{diff,amp}} = I_{FS} \cdot R_3 = 0.004 \text{ A} \times 249 \Omega = 0.996 \text{ V}.$$

The ADA4807-2 is powered from sources V3, V4: $\pm 2.5 \text{ V}$. Resistors R_3 and R_5 are chosen as 249 Ω from the standard series, ensuring high linearity of conversion at currents up to 4 mA.

ADA4870 Input Network

Resistors R_a and R_b

R_a (150 Ω in the current revision) is connected *in series* between the output of U2 (node N001) and the inverting input of ADA4870 (node OUTA). This node also carries the feedback resistor R_f — so R_a is in the low-impedance summing node circuit, through which real signal current flows (not a shunt to ground, as one might assume by analogy with a classic 4-resistor differential amplifier).

R_b (140 Ω) is connected *in series* between the output of U3 (node N002) and the non-inverting input of ADA4870 (node OUTB). There is no free shunt to ground in this topology on either input.

The formula

$$R_b = \frac{R_a \cdot R_f}{R_a + R_f} \quad (8)$$

remains valid, but its physical meaning is input bias current matching: it equalizes the DC resistance seen by each op-amp input, thereby removing the contribution of input bias current to output offset. **It does not nullify the DC component at the output** — contrary to what the name "symmetry condition" might suggest.

The DC component at the output arises from the very nature of the DAC currents: I_{OUTA} and I_{OUTB} are unidirectional, their sum $I_{OUTA} + I_{OUTB} = I_{FS} = \text{const}$, not zero. This common-mode current flows through the asymmetric (single feedback resistor on the inverting side) ADA4870 network and produces an output offset

$$V_{\text{offset}} = -\frac{R_{TIA} \cdot I_{FS}}{2}, \quad (9)$$

which **does not depend on the choice of** R_a , R_f , R_b — only on R_{TIA} and I_{FS} . With $R_{TIA} = 249\ \Omega$, $I_{FS} = 4\text{ mA}$: $V_{\text{offset}} \approx -0.498\text{ V}$ — this is exactly the $\sim -0.5\text{ V}$ observed in simulation, not a consequence of inaccurate E96 rounding of R_b (see also section 3.1 and the corrected note in section 5.2).

Separately from the offset — the gain asymmetry between the rails (see equation (11) and the comment after it): this is also structural, but a different effect not directly related to offset, and also not fixable by tuning R_a/R_b .

Important: swapping R_a and R_b would violate condition (8), because node OUTA additionally has the feedback resistor R_f . Therefore R_b must always equal $R_a \parallel R_f$, not the reverse.

DAC Parasitic Elements (R_{paraA}/C_{paraA} , R_{paraB}/C_{paraB})

These elements model the output impedance of the AD9705 current outputs. Typical values from the datasheet [3] are used:

- $R_{paraA} = R_{paraB} = 150\text{ k}\Omega$,
- $C_{paraA} = C_{paraB} = 5\text{ pF}$.

In parallel, capacitors $C_a = C_b = 22\text{ pF}$ are also included (model of parasitic board capacitance). Thus, total capacitance at each input node (INN and INP) is $5 + 22 = 27\text{ pF}$.

The influence at low frequencies (1 MHz) is negligible. The pole formed by 27 pF and resistance of about 100 Ω lies at about 59 MHz, above the operating band.

Feedback Resistors R_{fa} , R_{fb}

The value of 249 Ω for resistors R_{fa} and R_{fb} in the transimpedance amplifiers (TIA) on ADA4807-2 was not chosen arbitrarily, but based on a balance between several key requirements. Here is the rationale:

- The DAC full-scale current was set to $I_{FS} = 4\text{ mA}$. This current determines the signal amplitude at the TIA output.
- Target output voltage: To effectively drive the subsequent ADA4870 stage, which requires about 1 V differential to obtain $\pm 14\text{ V}$ amplitude, a value $V_{diff_amp} \approx 0.996\text{ V}$ was chosen.
- For a classic TIA with op-amp, where current flows into the virtual ground of the inverting input, the output voltage is simply: $V_{out} = I_{in} \times R_f$. In our case:

$$R_f = \frac{V_{diff_amp}}{I_{FS}} = \frac{0.996\text{ V}}{0.004\text{ A}} \approx 249\Omega \quad (10)$$

Thus, 249 Ω is a standard E96 series value that perfectly converts 4 mA current into about 1 V.

A different value could have been used, but 249 Ω proved optimal because:

It does not overload the DAC: The DAC output voltage (about 1 V) remains within its compliance range, ensuring linearity.

- It does not overload the ADA4807 input: The TIA output stays within the linear range of the op-amp with $\pm 2.5\text{ V}$ supply.
- It leaves gain margin: Even a small change in this resistor would require recalculation of R_a and R_b in the output stage to maintain balance.

If the DAC output current I_{FS} needs to be changed, this resistor should be recalculated using the same formula to keep the same differential voltage for the ADA4870 stage.

Amplifier U1 (ADA4870)

The ADA4870 is connected in a differential configuration. The signal from U2 output (node N001) is fed to the **inverting** input via resistor R_a (the same node OUTA carries the feedback resistor R_f), while the signal from U3 output (node N002) is fed to the **non-inverting** input via R_b . Feedback is provided by resistor R_f from the SIGNAL output to the inverting input (node OUTA).

The differential signal gain is:

$$A_v = \frac{R_f}{R_a}. \quad (11)$$

With $R_f = 1.4 \text{ k}\Omega$ and $R_a = 100 \text{ }\Omega$, $A_v = 14$.

Note on asymmetry. Formula (11) is the gain only for the inverting path. For the non-inverting path (signal via R_b), the gain is $1 + R_f/R_a$, which is exactly 1 larger in magnitude — this is a topological property of the single-feedback-resistor circuit, independent of the specific values. With target $A_v \approx 14$, the discrepancy (14 vs 15) gives an output peak asymmetry of about 6–7%, which is considered acceptable given the available voltage headroom (see also section 3.1).

The output voltage amplitude for the given DAC current:

$$V_{\text{peak}} = A_v \cdot V_{\text{diff,amp}} = \frac{R_f}{R_a} \cdot I_{FS} \cdot R_3. \quad (12)$$

For $I_{FS} = 4 \text{ mA}$, $R_3 = 249 \text{ }\Omega$, $R_a = 100 \text{ }\Omega$, $R_f = 1.4 \text{ k}\Omega$, we obtain:

$$V_{\text{peak}} = 14 \times 0.996 \approx 13.94 \text{ V}.$$

Peak load current for $R_{Load} = 22 \text{ }\Omega$:

$$I_{\text{peak}} = \frac{14 \text{ V}}{22 \text{ }\Omega} \approx 0.636 \text{ A} < 1 \text{ A}.$$

ADA4870 supply: $\pm 18 \text{ V}$ from sources V1, V2.

Feedback Network (R_f , C_f)

R_f (1.4 k Ω) is the main feedback resistor defining the gain.

C_f (2 pF) is connected in parallel with R_f . It introduces a zero in the loop transfer function, compensating the pole from the parasitic capacitance at the ASA input. The zero frequency:

$$f_z = \frac{1}{2\pi R_f C_f} = \frac{1}{2\pi \times 1400 \times 2 \times 10^{-12}} \approx 56.8 \text{ MHz}.$$

The optimal value is $C_f = C_{\text{INN}} \cdot R_a/R_f \approx 27 \text{ pF} \times 100/1400 \approx 1.9 \text{ pF}$. The used value of 2 pF is close to the calculated one and ensures stability.

Load Network (R_{Load} , C_{Load})

$R_{Load} = 22 \text{ }\Omega$ — active load resistance. Peak current 0.636 A, well below the maximum ADA4870 current (1 A).

$C_{Load} = 220 \text{ pF}$ — model of parasitic board and load capacitance. At 1 MHz its impedance is about 723 Ω , much larger than 22 Ω , so the shunting effect is minimal.

SD Divider (R1, R2)

The divider formed by R1 (2 k Ω) and R2 (10 k Ω) sets the voltage on the SD enable input of the ADA4870. With ± 18 V supply, the SD voltage is $-18 \text{ V} \times \frac{10}{12} = -15 \text{ V}$, which falls within the enable range (from -16.9 V to -13 V).

Theoretical Calculation

Component Selection Algorithm

Based on the given requirements (V_{out_amp} , R_{load} , I_{FS}) and component datasheets, an automated algorithm for selecting R_a , R_f , R_b , C_f has been developed. Key steps:

1. Feasibility check.

- Maximum undistorted amplitude: $V_{max} = |V_{sup}| - V_{headroom}$.
- Peak load current: $I_{peak} = V_{out_amp}/R_{load}$. If $I_{peak} > I_{out_max}$ or $V_{out_amp} > V_{max}$, calculation stops.

2. TIA differential voltage. $V_{diff_amp} = I_{FS} \cdot R_{TIA}$ (typically 0.996 V for 4 mA and 249 Ω).

3. Required gain. $A_v = V_{out_amp}/V_{diff_amp}$.

4. Iterate over R_a candidates. For each value from the list $R_{a_candidates}$:

$$\begin{aligned} R_f &= A_v \cdot R_a; \\ R_b &= \frac{R_a \cdot R_f}{R_a + R_f} \quad (\text{ideal balance condition}); \\ C_f &= C_{f_base} + \frac{\max(0, R_f - R_{f_threshold})}{1000} \cdot C_{f_per_kohm}. \end{aligned}$$

R_f and R_b are rounded to the nearest E96 (1%) series value. Variants with $R_f > R_{f_max}$ are discarded.

5. Ranking variants. All valid combinations are sorted by the criterion of **minimum balancing error of R_b** :

$$\delta R_b = |R_{b_ideal} - R_{b_E96}| \rightarrow \min.$$

The smaller δR_b , the lower the DC component at the ADA4870 output. In case of equal δR_b , preference is given to the combination where R_f is closest to the manufacturer recommended value (see Table 1).

Recommended R_f Values for ADA4870

The ADA4870 is a current-feedback amplifier (CFB), and the choice of R_f significantly affects bandwidth and stability. The manufacturer recommends optimal R_f for typical gains (see Table 6 in the datasheet [1]):

Табл. 1: Recommended R_f for ADA4870

Gain (V/V)	R_f (Ω)
+1	2000
-1	1210
+2	1500
-2	1210
+5, +10	1210

In our differential configuration, the effective gain exceeds 10, so the target guideline for R_f is **1.21 k Ω** . This guideline is used as a secondary sorting key.

Example Calculation for this Circuit

- $V_{\text{peak}} = 14 \text{ V}$, $R_{\text{Load}} = 22 \text{ } \Omega \Rightarrow I_{\text{peak}} = 0.636 \text{ A}$.
- $I_{FS} = 4 \text{ mA}$, $R_{TIA} = 249 \text{ } \Omega \Rightarrow V_{\text{diff,amp}} = 0.996 \text{ V}$.
- $R_a = 100 \text{ } \Omega$, required $A_v = 14/0.996 \approx 14.06$.
- $R_f = 14.06 \times 100 = 1406 \text{ } \Omega \rightarrow$ choose 1.4 k Ω (E96: 1.40) or 1.43 k Ω (closer). We take $R_f = 1400 \text{ } \Omega$, then $A_v = 14.00$.
- $R_b = (100 \times 1400)/(1500) = 93.33 \text{ } \Omega \rightarrow 93.1 \text{ } \Omega$ (E96).
- $C_f \approx (27 \times 10^{-12}) \times 100/1400 \approx 1.9 \text{ pF} \rightarrow$ keep 2 pF.
- Check: $V_{\text{peak}} = A_v \cdot V_{\text{diff,amp}} = 14 \times 0.996 = 13.94 \text{ V}$, meeting the requirement.

Final Component Values

Updated: the table below reflects the current algorithm selection (section 3.1) with current `Ra_candidates` and `Rf_target=1210` in the config — $R_a = 150 \text{ } \Omega$, $R_f = 2100 \text{ } \Omega$ (E96), $R_b = 140 \text{ } \Omega$ (E96, exact match to $R_a \parallel R_f$, $\delta R_b = 0$). The example in section 3.3 above (with $R_a = 100 \text{ } \Omega$) is an earlier iteration, kept for illustration of the calculation flow, but is not the current operating point.

Табл. 2: Component values of schematic ada4807_4870.asc (current revision)

Component	Value	Purpose
R_f	2.1 k Ω	Transimpedance, sets amplitude
R_a	150 Ω	Resistor to inverting input of U1
R_b	140 Ω	Resistor to non-inverting input of U1 ($R_a \parallel R_f$)
R_3, R_5	249 Ω	TIA resistors (current-to-voltage conversion)
C_f	1.9 pF	Feedback compensation
Ca, Cb	22 pF	Parasitic capacitance of DAC inputs
R_{paraA}, R_{paraB}	150 k Ω	DAC output resistance
C_{paraA}, C_{paraB}	5 pF	DAC output capacitance
C_{Load}	220 pF	Parasitic load capacitance
R_{Load}	22 Ω	Load
R1, R2	2 k Ω , 10 k Ω	SD voltage divider
V_{sup}	± 18 V	ADA4870 supply
$V_{sup,TIA}$	± 2.5 V	ADA4807-2 supply

Adaptive Simulation Time Setting

To obtain reliable Fourier analysis results at any frequency, it is necessary that an integer number of signal periods fits into the analysis window and that transients have settled. The program implements dynamic generation of the `.tran` directive based on parameters set in `config.yaml` (previously `config.json` before switching to YAML):

- `periods_transient = 10` – number of periods for transient decay;
- `periods_analysis = 10` – number of periods whose data are saved for analysis;
- `points_per_period = 1000` – minimum number of points per period.

From these values, the following are automatically computed:

$$T = \frac{1}{f_{\text{signal}}}, \quad (13)$$

$$t_{\text{end}} = (N_{\text{transient}} + N_{\text{analysis}}) \cdot T, \quad (14)$$

$$t_{\text{start}} = N_{\text{transient}} \cdot T, \quad (15)$$

$$\Delta t_{\text{max}} = \frac{T}{N_{\text{points}}}. \quad (16)$$

Thus, for low frequencies the simulation becomes longer, but guarantees that transients have settled before saving starts, and Fourier analysis is performed on an integer number of

periods. This eliminates spurious THD behavior and ensures a physically meaningful frequency dependence of distortion.

Thermal Conditions of ADA4870

The average power dissipated by the amplifier with a sinusoidal signal [1]:

$$P_{AVG} = V_S \cdot I_q + \frac{2}{\pi} \cdot \frac{V_{CC} V_{peak}}{R_L} - \frac{V_{peak}^2}{2R_L}, \quad (17)$$

where $V_S = 36$ V (total), $I_q = 32.5$ mA, $V_{CC} = 18$ V, $V_{peak} = 14$ V, $R_L = 22$ Ω :

$$P_{AVG} = (36 \times 0.0325) + \frac{2}{\pi} \cdot \frac{18 \times 14}{22} - \frac{14^2}{2 \times 22} = 1.17 + \frac{2}{\pi} \cdot 11.4545 - 4.4545 \approx 1.17 + 7.29 - 4.45 = 4.01 \text{ W}.$$

Junction temperature at ambient $T_{amb} = 40$ °C and case-to-ambient thermal resistance θ_{CA} :

$$T_J = T_{amb} + P_{AVG} \cdot (\theta_{JC} + \theta_{CA}),$$

where $\theta_{JC} = 1.3$ °C/W (for SOIC package). For a safe margin ($T_J \leq 150$ °C), we require

$$\theta_{CA} \leq \frac{150 - 40}{4.01} - 1.3 \approx 27.4 - 1.3 = 26.1 \text{ °C/W}.$$

A heatsink with thermal resistance not exceeding 20 °C/W is recommended.

Load power:

$$P_{R_{load}} = \frac{V_{peak}^2}{2R_{load}} = \frac{196}{44} \approx 4.45 \text{ W}.$$

The load resistor should be rated for at least 10 W.

LTspice Simulation Results

Simulation was performed in LTspice version 26.0.1 for Windows [5]. **Note:** the run below is an early iteration with $R_f = 820$ Ω (historical, for illustration of the methodology), giving an amplitude of about 8.65 V on a 22 Ω load, which is lower than the target 14 V. Current values and results are in section 6.1 and Table 2. Simulation directives:

- .tran 10us
- .FOUR 1Meg V(aout)
- .MEAS TRAN Vfinal FIND V(aout) AT=10us

Simulation Parameters

Табл. 3: Simulation parameters (.param)

Parameter	Value	Purpose
IFS	4 mA	DAC full-scale current
FREQ	1 MHz	Signal frequency
R_f Val	820 Ω	Feedback resistor (base configuration)
R_{Load} Val	22 Ω	Load resistance

Fourier Analysis of Output Signal ($R_f = 820 \Omega$)

Табл. 4: Fourier analysis of $V(aout)$, $f_0 = 1$ MHz (from .log file)

No.	Frequency, Hz	Amplitude, V	Norm.	Phase, $^\circ$	Norm. phase, $^\circ$
1	1.000×10^6	8.645×10^0	1.000	92.73	0.00
2	2.000×10^6	1.647×10^{-4}	1.905×10^{-5}	140.79	48.06
3	3.000×10^6	1.254×10^{-4}	1.450×10^{-5}	-69.62	-162.35
4	4.000×10^6	1.373×10^{-4}	1.588×10^{-5}	94.03	1.30
5	5.000×10^6	1.213×10^{-4}	1.403×10^{-5}	-127.85	-220.57
6	6.000×10^6	1.266×10^{-4}	1.465×10^{-5}	37.79	-54.94
7	7.000×10^6	1.439×10^{-4}	1.665×10^{-5}	179.05	86.32
8	8.000×10^6	1.378×10^{-4}	1.594×10^{-5}	-13.93	-106.65
9	9.000×10^6	1.650×10^{-4}	1.908×10^{-5}	137.08	44.35

Табл. 5: Simulation summary ($R_f = 820 \Omega$)

Parameter	Value
Fundamental amplitude	8.645 V
DC offset	-0.5446 V
Total THD	0.00633%
V_{aout} at $t = 10 \mu s$	-0.956 V
V_{SD}	-15.183 V

Corrected: the measured DC shift (-0.54 V) is not a consequence of rounding R_b to $88.7\ \Omega$ instead of the exact $93.33\ \Omega$ (this error contributes two orders of magnitude less). It is the structural offset $V_{\text{offset}} = -R_{TIA} \cdot I_{FS}/2 \approx -0.498$ V (see equation (9)) — it does not depend on R_b and is not eliminated by precise tuning of R_b , whether with $R_f = 820\ \Omega$ or $R_f = 1.4\ \text{k}\Omega$.

Expected Performance with $R_f = 1.4\ \text{k}\Omega$

With R_f increased to $1.4\ \text{k}\Omega$ and the circuit balanced, the following parameters are expected:

$$V_{\text{peak}} \approx 14\ \text{V}, \quad \text{DC offset} \approx -0.5\ \text{V}, \quad \text{THD} < 0.01\%.$$

Peak current into $22\ \Omega$ load will be about 0.64 A. The ADA4870 power dissipation will increase to ~ 4 W, requiring a heatsink.

Simulation Results

At the outputs I_{NN} and I_{NP} we obtain:

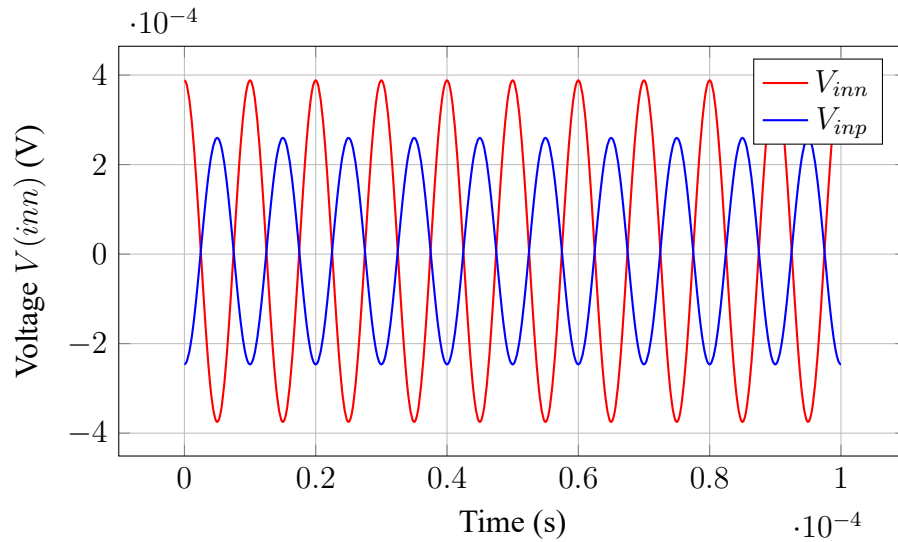


Рис. 3: Voltage at nodes I_{NN} , I_{NP}

Then, the output signal looks like:

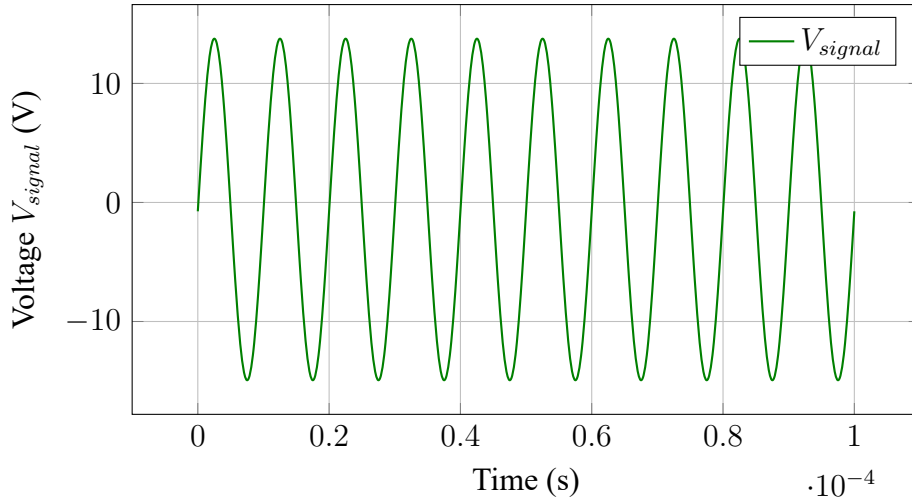


Рис. 4: ADA4870-1 output voltage V_{signal}

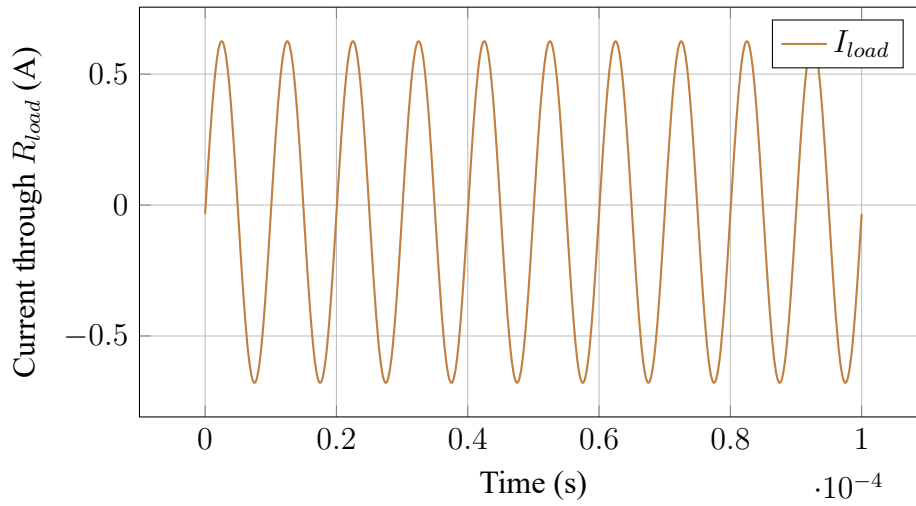


Рис. 5: Current through R_{load}

THD Degradation vs Frequency

To evaluate the operating frequency range and the behaviour of nonlinear distortion, the `degradation_sweep` method is used – sequential simulation runs at several frequencies with fixed component values. To avoid artefacts related to transients and insufficient analysis duration, adaptive simulation time settings are used (see section 3.5).

Figure 6 shows the THD vs frequency plot obtained in the range 100 kHz – 5 MHz for the selected component combination ($R_a = 150\ \Omega$, $R_f = 2100\ \Omega$, $R_b = 140\ \Omega$). **Corrected:** the dependence is not monotonic — THD rises from $\sim 0.060\%$ (100 kHz) to a peak of $\sim 0.063\%$ around 500 kHz–1 MHz, then decreases to $\sim 0.049\%$ at 5 MHz. This behaviour is atypical for simple reduction of feedback depth with frequency (which would give monotonic increase) — a possible reason is that C_f together with R_f creates a pole that suppresses higher harmonics more strongly at the upper end of the range than the amplifier's own distortion increases; this requires separate verification if the exact physical mechanism is needed.

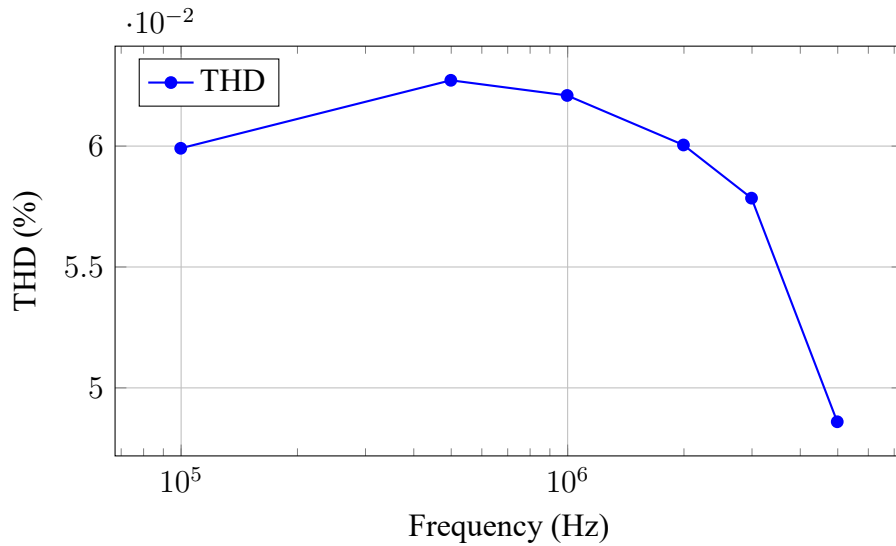


Рис. 6: THD degradation vs frequency. Data obtained via automated sweeping.

The obtained dependence confirms that throughout the range 100 kHz–5 MHz, distortion remains very low ($\text{THD} \lesssim 0.063\%$, with a small peak in the middle rather than monotonically increasing towards the upper bound) at an output power of ~ 4.5 W. The distortion margin for the target frequencies (up to 2 MHz per specification) is more than sufficient.

Output Protection Recommendations

ADA4870 Output Protection

When driving inductive loads, voltage spikes beyond the supply rails may occur. It is recommended to install Schottky protection diodes directly at the AOUT pin.

Connection diagram (Fig. 7):

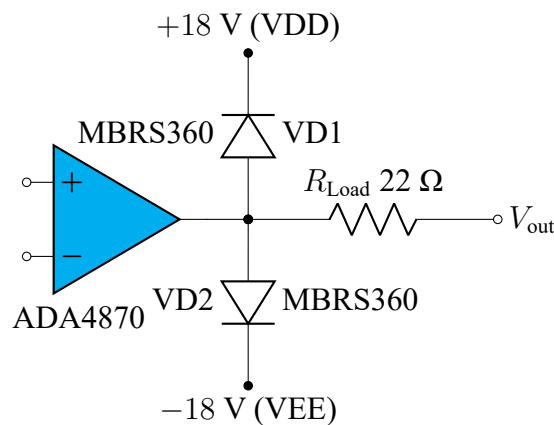


Рис. 7: Protection diode connection at ADA4870 output.

- **VD1** (to positive rail): anode to AOUT, cathode to +18 V (VDD).
- **VD2** (to negative rail): anode to −18 V (VEE), cathode to AOUT.

Diode Selection MBRS360

MBRS360 [4] is a 3 A, 60 V Schottky diode in SMC package. The peak load current of 0.636 A is well below the limit, reverse voltage 60 V safely covers possible spikes. The diode capacitance (about 180 pF) has negligible effect on signals up to 2 MHz.

Components for the Actual PCB

- R_{Load} : non-inductive resistor 22 Ω , rated for at least 10 W.
- **Protection diodes**: MBRS360 (or similar with current rating ≥ 1 A).
- **Bypass capacitors**: 100 nF + 10 μ F from each supply rail to GND, placed close to ADA4870.
- C_f : start with 2 pF, if necessary increase to 2.7–3.3 pF based on AC analysis.
- **TIA circuit**: resistors $R_3 = R_5 = 249 \Omega$, no capacitors in the ADA4807-2 feedback loop are required (driving a low-impedance load).

Список литературы

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